Department of Computer Science & Engineering

Birla Institute of Technology, Mesra, Ranchi - 835215

(India) Master of Computer Application

PROGRAMME COURSE STRUCTURE (ALL SEMESTERS)

BIRLA INSTITUTE OF TECHNOLOGY-MESRA, RANCHI

NEW COURSE STRUCTURE – Proposed from Monsoon 2020

Based on CBCS & OBE model

Recommended scheme of study for MCA Programme

SEMESTER 1: FIRST/MONSOON (Course Level: FOURTH)

Category of Course	Course Code	Courses	L	T	P	Total Credits
THEORY Programme Core (PC)	CA403	Computer Organization & Architecture	3	0	0	3
Programme Core (PC)	CA405	Data Structures and Algorithms	3	0	0	3
Programme Core (PC)	CA407	Database Design Concepts	3	0	0	3
Programme Core (PC)	CA409	Object Oriented Design using Java	3	0	0	3
Programme Core (PC)	CA411	Modern Operating Systems	3	0	0	3
LABORATORIES Programme Core (PC)	CA406	Data Structures and Algorithms Lab	0	0	3	1.5
Programme Core (PC)	CA408	Database Design Concepts Lab	0	0	3	1.5
Programme Core (PC)	CA410	Object Oriented Design using Java Lab	0	0	3	1.5
	MT132	Communication Skill-I	0	0	3	1.5
TOTAL						21

SEMESTER 2: SECOND/SPRING

Course Level	Category of Course	Course Code	Courses	L	T	P	C
FIRST	THEORY HSS	MT114	Fundamentals of management & Organization Behaviour	3	0	0	3

FOURTH	Programme Core (PC)	CA413	Data Communication & Computer Networks	3	0	0	0	3
FOURTH	Programme Core (PC)	CA415	Software Engineering Principles	3	0	0	0	3
FOURTH	Programme Core (PC)	CA417	Theory of Computation	3	0	0	0	3
FOURTH	Programme Core (PC)	CA419	Analysis of Algorithms	3	0	0	0	3
FOURTH	Program Elective	-	Program Elective-I	3	0	0	0	3
FOURTH	LABORATORIES Programme Core (PC)	CA414	DCCN Lab	0	0	0	3	1.5
FOURTH	Programme Core (PC)	CA416	Software Engineering Lab	0	0	0	3	1.5
FOURTH	Programme Core (PC)	CA422	IT Tools & Techniques Lab	0	0	0	3	1.5
FOURTH		MT133	Communication Skill-II	0	0	0	3	1.5
TOTAL								24

SEMESTER 3: THIRD/MONSOON (Course Level: FIFTH)

Category of Course	Course Code	Courses	L	T	P	Total Credits
THEORY Programme Core (PC)	CA511	Basics of Machine Learning	3	0	0	3
Programme Core (PC)	CA513	Compiler Design	3	1	0	4
Programme Core (PC)	CA515	Soft Computing	3	0	0	3
SIT / Soft Skill Course / Extra OE	CA550	Small Industrial Training/Small Project/MOOC				6
Program Elective	-	Program Elective-II	3	0	0	3
Program Elective	-	Program Elective-III	3	0	0	3
LABORATORIES Programme Core (PC)	CA512	Basics of Machine Learning Lab	0	0	3	1.5
Programme Core (PC)	CA514	Compiler Design Lab	0	0	3	1.5
TOTAL						25

SEMESTER 4: FOURTH/SPRING (Course Level: FIFTH)

Category of Course	Course Code	Courses	Total Credits
PROJECT Project	CA590	Project	20
TOTAL			20

TOTAL CREDITS: 90

LIST OF PROGRAM ELECTIVES

PE1 (Level 4)

Level	Course Code	Name of the PE courses	L	T	P	Credit
4	CA431	Distributed Databases Concepts	3	0	0	3
4	CA433	Intrusion Detection System	3	0	0	3
4	CA435	Modern Artificial Intelligence	3	0	0	3
4	CA437	Information Retrieval	3	0	0	3
4	CA439	Image Processing	3	0	0	3
4	CA441	Data Mining Techniques	3	0	0	3

PE2 (Level 5)

Level	Course Code	Name of the PE courses	L	T	P	Credit
5	CA519	Mobile Computing	3	0	0	3
5	CA521	Cyber Security	3	0	0	3
5	CA523	Cloud Computing	3	0	0	3
5	CA525	Deep Learning	3	0	0	3
5	CA527	Computer Vision	3	0	0	3
5	CA529	Network Security & Cryptography	3	0	0	3

PE2/PE3 (Level 5)

Level	Course Code	Name of the PE courses	L	T	P	Credit
5	CA539	Parallel Computing	3	0	0	3
5	CA541	Digital Forensic	3	0	0	3
5	CA543	Internet of Things (IOT)	3	0	0	3
5	CA545	Natural Language Processing	3	0	0	3
5	CA547	Big Data Analytics	3	0	0	3
5	CA549	Block Chain Technology	3	0	0	3